



HWA CHONG INSTITUTION
2018 JC2 PRELIMINARY EXAMINATION

MATHEMATICS
Higher 2

9758/02

Paper 2

Monday

17 September 2018

3 hours

Additional materials: Answer paper
 List of Formula (MF26)
 Cover Page

READ THESE INSTRUCTIONS FIRST

Write your name and CT class on all the work you hand in, including the Cover Page.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Do not write anything on the List of Formula (MF26).

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use a graphing calculator.

Unsupported answers from a graphing calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphing calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question. At the end of the examination, place the completed cover page on top of your answer scripts and fasten all your work securely together with the string provided.

This question paper consists of 6 printed pages.

Section A: Pure Mathematics [40 marks]

1 (a) Find $\int \frac{x+1}{x^2+3x+9} dx$. [4]

(b) Use the substitution $x = \cos \theta$ to find the exact value of $\int_0^{\frac{1}{\sqrt{2}}} \frac{x^2}{\sqrt{1-x^2}} dx$. [4]

2 (i) Using the standard series from the list of Formulae (MF26), show that the series expansion for $\frac{1}{1+\sin x}$ can be approximated to $1-x+x^2-\frac{5}{6}x^3$ when x is sufficiently small. [3]

(ii) Let $f(x) = \frac{1}{1+\sin x^3}$. Using the expansion in part **(i)**, find $f^{(9)}(0)$. [2]

(iii) “The series expansion of $\left(1+\frac{x}{a}\right)^b$ is equal to the series expansion of $\frac{1}{1-\sin x}$ as far as the term in x^3 where a and b are constants.”

Write down the series expansion of $\left(1+\frac{x}{a}\right)^b$ and $\frac{1}{1-\sin x}$ as far as the term in x^3 .

Hence justify if the above statement is valid. [4]

3 The function f is defined as

$$f : x \mapsto \begin{cases} ae^{a-x} & \text{for } 0 \leq x < a, \\ \left| \frac{1}{a}(x-a)^2 - a \right| & \text{for } x \geq a. \end{cases}$$

(i) Sketch the graph of $y = f(x)$, indicating clearly the axial intercepts. Show that f^{-1} does not exist. [4]

(ii) If the domain of f is restricted to $[0, k]$, determine the largest value of k in terms of a such that f^{-1} exists. [1]

Use the domain found in part **(ii)** for the rest of the question.

(iii) Define f^{-1} in similar form. [4]

(iv) Show that $ff\left(\frac{a}{2}\right) = ae^{\frac{a}{2}}\left(2-e^{\frac{a}{2}}\right)$ where $0 < a \leq \ln 2$. [2]

- 4 A line l has equation $x-1=\frac{y}{2}=z-3$, and a plane p has equation

$$\mathbf{r} = \begin{pmatrix} 5 \\ -2 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ -3 \end{pmatrix} + \mu \begin{pmatrix} k \\ 1 \\ 4 \end{pmatrix},$$

where k is a real constant, $\lambda, \mu \in \mathbb{R}$.

- (i) Given that l and p are parallel, show that $k = -1$. [3]

Use $k = -1$ for the rest of the parts of the question.

- (ii) Hence show that l and p do not intersect. [2]

- (iii) Find the exact distance between l and p . [3]

- (iv) A point A on l has coordinates $(2, 2, 4)$ and N is the foot of the perpendicular from A to p . Find the coordinates of N . Hence find the coordinates of the reflection of N in l . [4]

Section B: Statistics [60 marks]

- 5 A discrete random variable Y takes non-negative integer values with probabilities given as follows:

y	0	1	2	...	n	...
$P(Y = y)$	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	...	$\frac{1}{2^{n+1}}$	...

- (i) Find the probability that Y is odd. [2]
- (ii) Y_1 and Y_2 are two independent observations of Y . Find the probability that the sum of Y_1 and Y_2 is less than 4, given that their sum exceeds 2. [4]

- 6 Six married couples are to be seated in a row at a concert. Find the number of ways they can sit if

- (i) each couple is to sit together, [2]
- (ii) all women are next to one another and all men are next to one another, such that no man can sit next to his wife. [3]

After the concert, one particular married couple leaves. The rest go to a restaurant where they sit at a round table. Find the probability that each man sits next to his wife, and men and women alternate. [2]

[Turn over]

- 7 An engineering team from a car manufacturer wants to test their cars' braking system. The car travels along a stretch of road with speed v km/h. When the brakes are applied, the car comes to rest after travelling a further distance of d metres. A random sample of 6 pairs of values of v and d collected by a trainee mechanic from the engineering team is shown below.

v	30	40	50	60	70	80
d	5.00	5.30	6.45	8.50	17.00	25.90

- (i) Draw a scatter diagram for these values, labelling the axes clearly. [2]

It is thought that the distance travelled d can be modelled by one of the following models.

$$\text{Model I : } d = av + b \text{ or}$$

$$\text{Model II : } d = e^{pv+q}$$

where a , b , p and q are constants.

- (ii) Find the value of the product moment correlation coefficient between
- (a) v and d ,
- (b) v and $\ln d$. [2]
- (iii) The trainee mechanic proposed that Model II is a better model than Model I. Use your answers to parts (i) and (ii) to explain why the trainee mechanic is right. [2]
- (iv) Find the equation of the regression line of $\ln d$ on v . [1]
- (v) Using the regression line in part (iv), find the value of v if the driver applies his brakes immediately upon seeing an obstacle that is 10 metres away and stops just in time before crashing into it. [1]
- (vi) The original data set contains 7 pairs of data with regression line $d = 0.4256v - 11.74$. The trainee mechanic found that he does not have the value of d when $v = 75$ from his record. Find the missing value of d correct to 2 decimal places. [3]

[Turn over

- 8** Mrs Lee claimed that the mean time taken by students to finish a meal during recess is not more than 20 minutes. Two students, Jack and Jill, decided to work together to test if Mrs Lee's claim is true. A total of 50 students were selected. The time, x minutes, by each of the 50 students to finish a meal during recess was recorded. The results are summarised by

$$\sum x = 1380, \quad \sum x^2 = 83000.$$

- (i) Find unbiased estimates of the population mean and variance of the time taken by a student to finish a meal during recess. [2]
 - (ii) Stating a necessary assumption, carry out a test of Mrs Lee's claim at the 5% level of significance. [5]
 - (iii) Explain, in the context of the question, the meaning of "at the 5% level of significance". [1]
 - (iv) Jack and Jill have just learnt hypothesis testing. Jack carried out the test as in part (ii) while Jill performed a 2-tail test at 5% level of significance. Without performing any further test, explain whether Jill has the same conclusion as Jack. [2]
- 9** A bag contains 2 red balls, 3 yellow balls and 1 blue ball. Sue and Ben play a game where each takes turns to draw a ball from the bag, with replacement. The number of red balls obtained in n fixed draws from the bag is denoted by R .
- (i) State, in context, an assumption satisfied by R for it to be well modelled by a binomial distribution. [1]
 - (ii) Sue and Ben each draws n times from the bag. Find the least n such that the probability of both getting a total of at most 10 red balls is not more than 0.5. [3]
 - (iii) Sue and Ben each draws 5 times from the bag. The player with more red balls drawn wins. Otherwise, the game ends in a draw. Find the probability that Sue wins the game if she draws more than 3 red balls. [3]

In a variation of the game, Sue draws balls at random from the bag, one at a time without replacement, and stops when she obtains 2 yellow balls. The total number of balls Sue has to draw from the bag before she stops is denoted by T . Find $E(T)$ and $\text{Var}(T)$. [5]

[Turn over]

- 10** Each morning, Tony drives from his home to his office and has to pass 4 traffic lights on his way. He has to reach his office by 8.50 am. The driving time to his office and the time held up at a traffic light junction, in minutes, may be assumed to follow normal distributions with means and standard deviations as summarized below:

	Mean	Standard deviation
Driving time	14	2.1
Time held up at a traffic light junction	μ	0.2

Tony leaves home at 8.30 am. If the probability that Tony is not late is 0.713, show that $\mu = 1.2$, correct to 1 decimal place. State an assumption needed in your calculations. [4]

- (i) For 10 mornings, Tony leaves home at 8.30 am. Find the probability that he arrives late at his office for the third time on the 10th day. [2]
- (ii) Find the probability that Tony's driving time to his office is less than 10 times the time he is held up at a traffic light junction. [2]
- (iii) Find the probability that Tony's driving time to his office and the total time he is held up at the 4 traffic light junctions differs by more than 8 minutes. [3]

Assume μ is unknown.

- (iv) The time held up at a traffic light junction is recorded on n randomly chosen occasions. Find the smallest n so that it is at least 98% certain that the sample mean time Tony is held up at a traffic light junction is within 5 seconds of μ . [3]